§11.3 The dot product

$\mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ $\quad \vec{u}\cdot\vec{v} = (u_1,u_2)\cdot(v_1,v_2) = u_1v_1 + u_2v_2$

$\mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}$ $\quad \vec{u}\cdot\vec{v} = (u_1,u_2,u_3)\cdot(v_1,v_2,v_3) = u_1v_1+u_2v_2+u_3v_3$

$(1,2,3)\cdot(4,-1,-1) = 4-2-3 = -1$

$\|\vec{u}\| = \sqrt{\vec{u}\cdot\vec{u}} = \sqrt{u_1^2+u_2^2+u_3^2}$

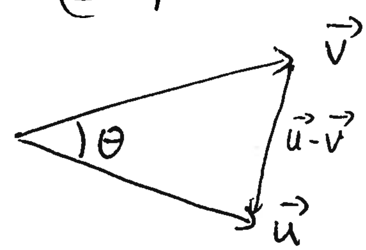
<u>Rules</u> ① $\vec{u}\cdot\vec{v} = \vec{v}\cdot\vec{u}$ ② $\vec{u}\cdot(\vec{v}+\vec{w}) = \vec{u}\cdot\vec{v}+\vec{u}\cdot\vec{w}$

③ $c(\vec{u}\cdot\vec{v}) = (c\vec{u})\cdot\vec{v}$ ④ $\vec{0}\cdot\vec{u} = 0$ ⑤ $\vec{u}\cdot\vec{u} = \|\vec{u}\|^2$

pf ① $\vec{u}\cdot\vec{v} = u_1v_1+u_2v_2+u_3v_3 = v_1u_1+v_2u_2+v_3u_3 = \vec{v}\cdot\vec{u}$

③ $c(\vec{u}\cdot\vec{v}) = c\circ(u_1v_1+u_2v_2+u_3v_3) = (cu_1)\cdot v_1 + (cu_2)v_2 + (cu_3)v_3$
$= (c\vec{u})\cdot\vec{v}$

<u>Thm</u> $\vec{u}\cdot\vec{v} = \|\vec{u}\|\,\|\vec{v}\| \cos\theta$



<u>Proof</u> Law of cosines says

① $\|\vec{u}-\vec{v}\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2 - 2\|\vec{u}\|\,\|\vec{v}\|\cos\theta$

② $\|\vec{u}-\vec{v}\|^2 = (\vec{u}-\vec{v})(\vec{u}-\vec{v}) = \vec{u}\cdot\vec{u} - \vec{v}\cdot\vec{u} - \vec{u}\cdot\vec{v} - \vec{v}\cdot\vec{v} = \|\vec{u}\|^2 + \|\vec{v}\|^2 - 2\vec{u}\cdot\vec{v}$

So $-2\|\vec{u}\|\,\|\vec{v}\|\cos\theta = -2\vec{u}\cdot\vec{v}$ so $\vec{u}\cdot\vec{v} = \|\vec{u}\|\,\|\vec{v}\|\cos\theta$

$\underline{Ex}$ Find angle between $(1,2)$ & $(2,1)$

$4 = 2+2 = \vec{u}\cdot\vec{v} = \|\vec{u}\|\,\|\vec{v}\|\cos\theta = \sqrt{5}\cdot\sqrt{5}\cos\theta$

$\cos\theta = 4/5 \qquad \theta = \cos^{-1}(0.8)$

$\underline{Ex}$ $\vec{u}$ & $\vec{v}$ are perpendicular if $\vec{u}\cdot\vec{v} = 0$
(because $\cos 90^\circ = 0$)

$\underline{Ex}$ Find a unit vector perpendicular (orthogonal) to $(3,4)$.

$(-4,3)$ is orthogonal since $(-4,3)(3,4) = -12+12 = 0$

$\|(-4,3)\| = \sqrt{16+9} = 5$ so $\left(-\frac{4}{5}, \frac{3}{5}\right)$ is the required unit vector.

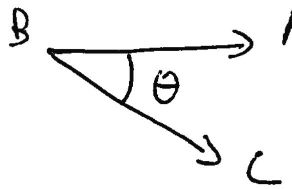
$\underline{Ex}$ for what t is $\begin{pmatrix} 1+t \\ -t \\ 2 \end{pmatrix}$ perpendicular to $\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$?

$\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1+t \\ -t \\ 2 \end{pmatrix} = 1+t-2t+2 = 3-t = 0$ if $t = 3$

the vector is $\begin{pmatrix} 4 \\ -3 \\ 2 \end{pmatrix}$

Ex compute the angle $\angle ABC$ where

$A = (2,1,0) \qquad B = (1,1,0) \qquad C = (3,2,1)$



$\vec{BA} = (2,1,0) - (1,1,0) = (1,0,0)$

$\vec{BC} = (3,2,1) - (1,1,0) = (2,1,1)$

$$2 = \vec{BA} \cdot \vec{BC} = \|\vec{BA}\| \, \|\vec{BC}\| \cos\theta = 1 \cdot \sqrt{6} \cdot \cos\theta$$

$$\cos\theta = 2/\sqrt{6} \qquad \theta = \cos^{-1}\left(2/\sqrt{6}\right)$$

Ex Find angles of $(1,2,3)$ with respect to x, y and z axis.

x axis $\quad i = (1,0,0)$

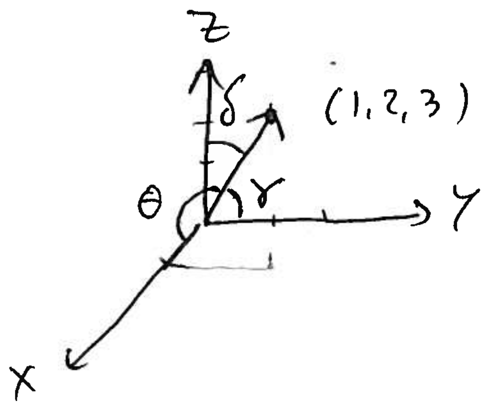
$$1 = (1,0,0) \cdot (1,2,3) = \|(1,0,0)\| \, \|(1,2,3)\| \cos\theta$$
$$= 1 \cdot \sqrt{14} \cos\theta$$

$$\cos\theta = 1/\sqrt{14} \qquad \theta = \cos^{-1}\left(1/\sqrt{14}\right)$$

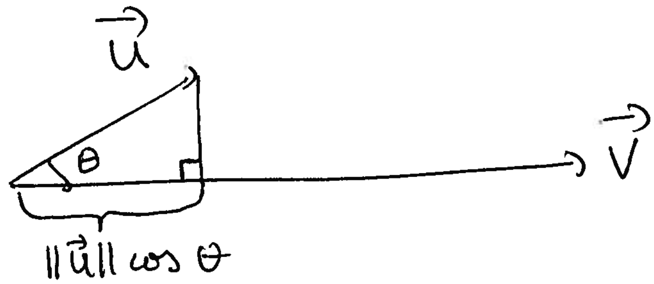
y-axis $\quad j = (0,1,0)$

$$2 = (0,1,0)(1,2,3) = \|(0,1,0)\| \, \|(1,2,3)\| \cos\gamma$$
$$= 1 \cdot \sqrt{14} \cos\gamma$$
$$\gamma = \cos^{-1}\left(2/\sqrt{14}\right)$$

z-axis $\quad j = (0,0,1) \qquad \delta = \cos^{-1}\left(3/\sqrt{14}\right)$



Projection of $\vec{u}$ on to $\vec{v}$



the projection is $pr_{\vec{v}}(\vec{u}) = \left(\frac{\vec{u}\cdot\vec{v}}{\|\vec{v}\|^2}\right)\cdot\vec{v}$

(or $\frac{\vec{u}\cdot\vec{v}}{\|\vec{v}\|}\cdot\frac{\vec{v}}{\|\vec{v}\|}$ since $\frac{\vec{u}\cdot\vec{v}}{\|\vec{v}\|} = \|\vec{u}\|\cos\theta$ &
$\frac{\vec{v}}{\|\vec{v}\|}$ is a unit vector)

Eg $\vec{v} = (1,0,0)$ $\quad \vec{u} = (3,1,5)$

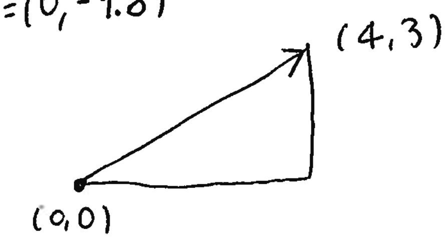
$pr_{\vec{v}}\vec{u} = \frac{3}{1}\cdot(1,0,0) = (3,0,0)$

$pr_{\vec{j}}\vec{u} = \frac{1}{1}\cdot\vec{j} = (0,1,0)$ $\qquad pr_{\vec{k}}\vec{u} = \frac{5}{1}\cdot\vec{k} = (0,0,5)$

Eg $\vec{v} = (1,2,3)$ $\quad pr_{\vec{v}}\vec{u} = \frac{3+2+15}{1+4+9}\cdot(1,2,3) = \frac{20}{14}(1,2,3)$
$= \left(\frac{10}{7}, \frac{20}{7}, \frac{30}{7}\right)$

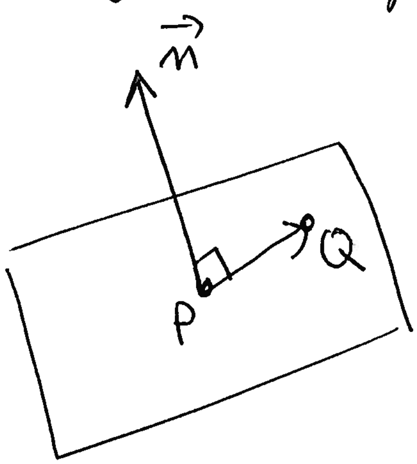
I physics $\quad$ Work $= \overrightarrow{\text{Force}}\cdot\overrightarrow{\text{Direction}}$

$F = (0, -9.8)$



work $= (0,-9.8)\cdot(4,3) = -29.4$

Equation of a plane perpendicular to $\vec{n} = (A, B, C)$ passing through point $P = (x_1, y_1, z_1)$



If $Q = (x, y, z)$

then $\vec{n} \cdot \vec{PQ} = 0$

so $(A, B, C) \cdot (x - x_1, y - y_1, z - z_1) = 0$

$$A(x - x_1) + B(y - y_1) + C(z - z_1) = 0$$

$$\text{So } Ax + By + Cz = \underbrace{Ax_1 + By_1 + Cz_1}_{D}$$

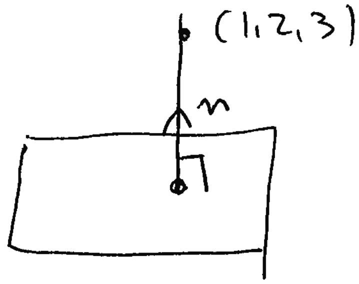
Eg $\vec{n} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ $P = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$

$$(1, 2, 3)(x - 2, y + 2, z - 1) = 0$$

$$x - 2 + 2y + 4 + 3z - 3 = 0$$

$$x + 2y + 3z = 1$$

Example Find the distance between $(1,2,3)$ and plane $x-2y+z=3$



Step 1 write line through $(1,2,3)$, orthogonal to plane

$$(1,2,3)+t(1,-2,1)$$

$\boxed{\text{Note } (1,-2,1) \text{ is } \vec{n}}$

Step 2 Intesect plane & line

$x = 1+t$

$y = 2-2t$

$z = 3+t$

$$1+t-2(2-2t)+3+t=3$$

$$-2t=3 \qquad t=-3/2$$

$$x=-1/2 \quad y=5 \quad z=3/2$$

Step 3 Find distance

$$\sqrt{(1+1/2)^2+(2-5)^2+(3-3/2)^2} = \sqrt{\frac{9}{4}+9+\frac{9}{4}} = \sqrt{\frac{27}{2}}$$

Ex For which t are $(1+2t, 2-t, 1+3t)$ and $(3, 1, -1)$ orthogonal?

Ex Find cosine of angle between $(1, 3)$ and $(2, 5)$ (and between $(1,1,1)$ and $(3,4,5)$)

Ex Show that $(1,1,1)$ and $(1,0,-1)$ are perpendicular. Find a third vector perpendicular to $(1,1,1)$ & $(1,0,-1)$.

Ex Find $\angle ABC$ where $A=(1,2,3)$ $B=(-4,5,6)$ $C=(1,0,1)$

Ex Find $pr_{\vec{v}}\vec{u}$ where $\vec{u}=(3,2,1)$ & $\vec{v}=(2,0,-1)$ (and $\vec{u}=(1,2,0)$ $\vec{v}=(2,0,3)$)

Ex Find the equation of a plane through $(-1, 2, -3)$ & parallel to the plane $2X+4Y-Z=6$

Ex Find the distance from $(2, 6, 3)$ to the plane $-3X+2Y+Z=9$

Ex Find the equation of the plane whose points are equidistant from $(-2,1,4)$ & $(6,1,-2)$